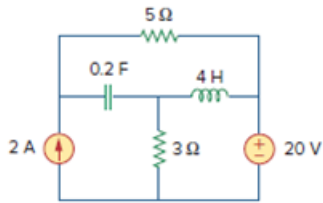


Problem

For the circuit in Fig. 8.50, obtain the dual circuit.

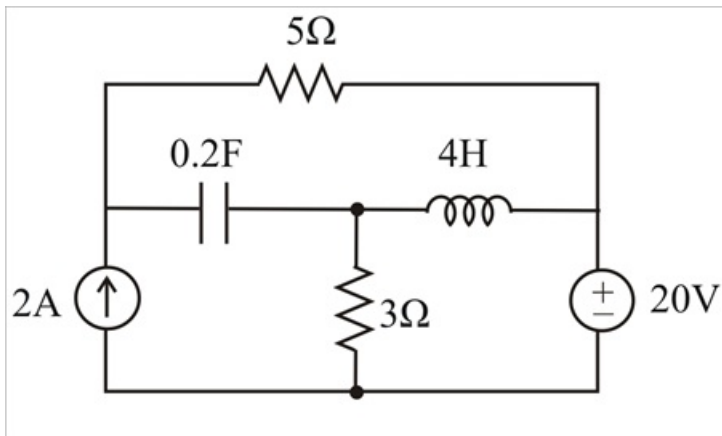
Figure 8.50:



Step-by-step solution

Step 1 of 4

The given circuit is



Step 2 of 4

We first locate nodes 1, 2 and 3 in the three meshes and also the reference node 0 for the dual circuit. Draw a line between one node and another crossing an element.

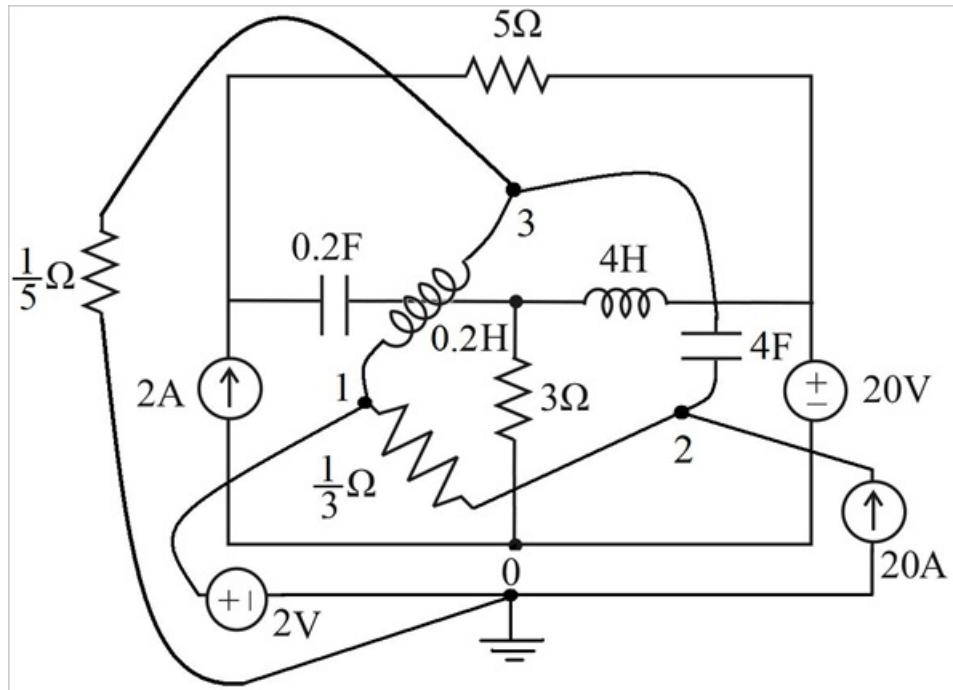
We replace the line joining the nodes by the duals of the elements which it crosses.

For example, the line between node 1 and 2 crosses 3Ω resistor, therefore it is replaced with a $\frac{1}{3}\Omega$ resistor.

The line between node 1 and 3 crosses 0.2F capacitor, therefore it is replaced with a 0.2H inductor.

We keep doing this until all the elements are crossed. The dual circuit is constructed on the original circuit as shown.

Step 3 of 4



Step 4 of 4

Hence, the dual circuit is

